

Series $\rightarrow G_1(s) \cdot G_2(s)$

parallel $\rightarrow G_1(s) + G_2(s)$

Summing point ahead $\rightarrow Q(s) = Z(s) \cdot G(s)$

Summing point behind $\rightarrow Q(s) = \frac{Z(s)}{G(s)}$

pick up point ahead $\rightarrow 1/G(s)$

pick up point behind $\rightarrow G(s)$

Reducing feed back loop $\rightarrow \frac{Y(s)}{X(s)} = \frac{G(s)}{1 + G(s)H(s)}$

$$G(s) = \frac{s+2}{(s+5)(s+1)} \rightarrow \frac{P(s)}{Q(s)}$$

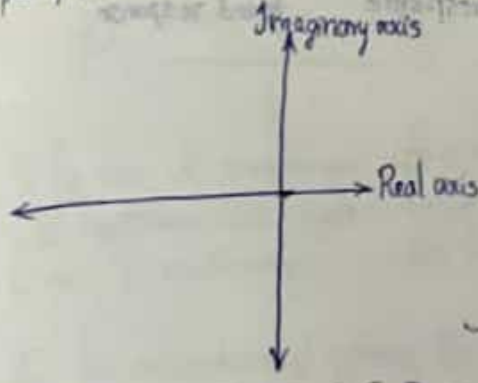
$$G(s) = K \frac{P(s)}{Q(s)}$$

$Q(s) = 0 \rightarrow$ characteristic eqn

$P(s) = 0 \rightarrow$ soln $\rightarrow$ zeroes $\rightarrow 0$

soln $\rightarrow$ poles $\rightarrow x$

$$s = \sigma + j\omega$$



pole-zero map

poles are zeroes of a system.

influence transient and steady state response

$$G(s) = K \frac{\prod (s+z_i)}{\prod (s+p_i)}$$

$K \rightarrow$ gain constant

$z_i \rightarrow$ zeroes

$p_i \rightarrow$ poles

$\prod \rightarrow$ product

1st Order System:

$$G(s) = \frac{a}{s+a} = \frac{1}{1+Ts}$$

Where, $T = \frac{1}{a}$

Step Response: Unit

$x(t) = 1 \quad X(s) = \frac{1}{s}$

$$Y(s) = \frac{1}{s} \left(\frac{a}{s+a} \right)$$

$$y(t) = 1 - e^{-at}$$

Error, $e(t) = \text{Desired output} - \text{Actual output} = 1 - (1 - e^{-at}) = e^{-at}$

$T \rightarrow$ time constant @ 63.2% of final output } Represents speed of response of system.

Smaller 'T' $\rightarrow$ Faster the response of system

when $t = T$
 $y(t) = 0.632$

Steady state (final) response $\rightarrow \lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s) = 1$

Steady state error $\rightarrow \lim_{t \rightarrow \infty} e(t) = 0$

Ramp Response: Unit

$x(t) = t \quad X(s) = \frac{1}{s^2}$

$$Y(s) = \frac{1}{s^2} \cdot \frac{a}{s+a}$$

$$\rightarrow y(t) = t - \frac{1}{a} (1 - e^{-at})$$

$$e(t) = \frac{1}{a} (1 - e^{-at})$$

steady state error $\rightarrow \frac{1}{a}$ (as $\lim_{t \rightarrow \infty} e(t)$)

Impulse Response: Unit

$x(t) = \delta(t) \quad X(s) = 1$

$$y(t) = a e^{-at}$$

$$R \rightarrow S \rightarrow I$$

IInd Order Systems

Second order, Ordinary, Non-homogeneous, linear differential eqn.

Spring, Mass, Dashpot system

$$m\ddot{x} + b\dot{x} + kx = F(t)$$

Homogeneous equation:

$$m\ddot{x} + b\dot{x} + kx = 0 \quad \text{Linear} \Rightarrow \text{any exp. fn is a possible soln.}$$

$$x(t) = Ae^{pt} \quad A \rightarrow \text{Arbitrary constant evaluated from initial condition.}$$

$p \rightarrow$ Constant (Not arbitrary)

$$\Rightarrow mp^2 Ae^{pt} + bp Ae^{pt} + k Ae^{pt} = 0$$

$$\Rightarrow Ae^{pt} (mp^2 + bp + k) = 0$$

$$\Rightarrow mp^2 + bp + k = 0 \quad \text{Characteristic eqn of system}$$

$$p = \frac{-b \pm \sqrt{b^2 - 4km}}{2m}$$

$$\sqrt{b^2 - 4km} \quad \text{Discriminant}$$

$$\text{Say, } x(t) = A_1 e^{p_1 t} + A_2 e^{p_2 t} \quad \left. \begin{array}{l} \text{Two soln of 2nd order diff eqn} \\ \text{By Super position principle} \end{array} \right\} \quad p_1 \text{ and } p_2 \text{ are roots}$$

If p_1 and p_2 are distinct $\rightarrow$ Real or complex

$p_1 = p_2 \rightarrow$ Real and equal

• Case I: $b^2 > 4km$

Roots are real and distinct.

Over damped system

Damping effect predominant

• Case II: $b^2 < 4km$

Roots are complex conjugate

Underdamped system

Damping effect less predominant.

• Case III: $b^2 = 4km$

Roots are real and equal

Critically damped system

$$b_c = \sqrt{4km} = 2m \sqrt{\frac{k}{m}}$$

$$b_c = 2m \omega_n$$

$$\omega_n = \sqrt{\frac{k}{m}}$$

$\omega_n \rightarrow$ Natural frequency of damped system

$$\frac{b}{b_c} = \zeta \quad (\text{Zeta}) \text{ Damping ratio} \Rightarrow b = \zeta \cdot 2m \omega_n$$

$\zeta = 1 \rightarrow$ Critically damped

$\zeta < 1 \rightarrow$ Under damped

$\zeta > 1 \rightarrow$ Over damped

$$p_{1,2} = \frac{-b}{2m} \pm \frac{\sqrt{b^2 - 4km}}{2m} = -\zeta \omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

(i) Overdamped ($\zeta > 1$)

p_1 and p_2 are real & distinct

$$x(t) = e^{-\zeta \omega_n t} (A_1 e^{\omega_n \sqrt{\zeta^2 - 1} t} + A_2 e^{-\omega_n \sqrt{\zeta^2 - 1} t})$$

(ii) Under damped ($\zeta < 1$)

p_1 and p_2 are complex conjugates

$$x(t) =$$

$$p_{1,2} = -\zeta \omega_n \pm j \omega_n \sqrt{1 - \zeta^2}$$

$$p_{1,2} = -\zeta \omega_n \pm j \omega_n \sqrt{1 - \zeta^2}$$

(iii) Critically damped ($\zeta = 1$)

p_1 and p_2 are real and equal

$$x(t) = (A_1 + A_2 t) e^{-\omega_n t}$$

$$\begin{aligned}
 x(t) &= e^{-\zeta \omega_n t} (A_1 e^{i\omega_d t} + A_2 e^{-i\omega_d t}) \\
 &= e^{-\zeta \omega_n t} (A_1 (\cos \omega_d t + i \sin \omega_d t) + A_2 (\cos \omega_d t - i \sin \omega_d t)) \\
 &= e^{-\zeta \omega_n t} ((A_1 + A_2) \cos \omega_d t + (A_1 - A_2) i \sin \omega_d t) \\
 &= e^{-\zeta \omega_n t} (C_1 \cos \omega_d t + C_2 \sin \omega_d t) \\
 &= e^{-\zeta \omega_n t} (X_0 \sin \theta (\cos \omega_d t + \cos \theta \sin \omega_d t)) \\
 &= X_0 e^{-\zeta \omega_n t} (\sin(\omega_d t + \theta))
 \end{aligned}$$

C_1 & $C_2 \rightarrow$ New arbitrary constants

$$\begin{aligned}
 C_1 &= X_0 \sin \theta \\
 C_2 &= X_0 \cos \theta \\
 X_0 &= \sqrt{C_1^2 + C_2^2} \\
 \theta &= \tan^{-1} \left(\frac{C_1}{C_2} \right)
 \end{aligned}$$

X_0 and θ are the new arbitrary constants

In the case of overdamped and critically damped system, one vibration is initiated by giving some initial conditions, the system comes to equilibrium without oscillation. The time taken by the system to come to equilibrium is smaller for critically damped system than an overdamped.

For the underdamped system, the response is a sine curve with amplitude $X_0 e^{-\zeta \omega_n t}$ and circular frequency ω_d .
Period of oscillation: $T = \frac{2\pi}{\omega_d}$. The amplitude of oscillation decays exponentially to zero as $t \rightarrow \infty$.

Only an underdamped system will vibrate/oscillate when it is disturbed from equilibrium.

"Root Locus"

In other way, $m\ddot{x} + b\dot{x} + kx = F(t)$

Taking Laplace transform,

$$X(s) = \frac{(ms+b)x(0) + m\dot{x}(0) + F(s)}{ms^2 + bs + k} = \underbrace{\frac{(ms+b)x(0) + m\dot{x}(0)}{ms^2 + bs + k}}_{\text{Natural response}} + \underbrace{\frac{F(s)}{ms^2 + bs + k}}_{\text{Forced response}}$$

Assuming no input, $F(s) = 0$, $x(0) = 1$ & $\dot{x}(0) = 0$

$$X(s) = \frac{ms+b}{ms^2 + bs + k} = \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

(i) $\zeta = 1$

(ii) $\zeta < 1$

(iii) $\zeta > 1$

$A_1 = 1$

$$X_0 = \frac{1}{\sqrt{1-\zeta^2}}$$

$A_2 = \omega_n$

$$C_1 = X_0 \sin \theta = \sqrt{1-\zeta^2}$$

$$C_2 = X_0 \cos \theta = \zeta$$

$$\tan \theta = \frac{\sqrt{1-\zeta^2}}{\zeta}$$

$$A_1 = \frac{\zeta + \sqrt{\zeta^2 - 1}}{2\sqrt{\zeta^2 - 1}}$$

$$A_2 = \frac{-\zeta - \sqrt{\zeta^2 - 1}}{2\sqrt{\zeta^2 - 1}}$$

If we set all initial condition to zero, $X(s) = \frac{F(s)}{ms^2 + bs + k}$

$$\Rightarrow G(s) = \frac{X(s)}{Y(s)} = \frac{k}{kms^2 + bs + k} = \frac{1}{k} \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = k \frac{P(s)}{Q(s)}$$

Standard form of 2nd order System: $\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

$q(s) = 0 \rightarrow$ Characteristic eqn

Roots of $q(s) \rightarrow$ poles

$$s = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

• Impulse: Unit $f(t) = \delta(t)$ $F(s) = 1$

$$X(s) = \frac{F(s)}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s + \zeta\omega_n)^2 - \omega_n^2 \sqrt{\zeta^2 - 1}}$$

(i) $\zeta = 1$

$$x(t) = \omega_n^2 t e^{-\omega_n t}$$

(ii) $\zeta > 1$

$$x(t) = e^{-\zeta\omega_n t} \left[\frac{\omega_n}{2\sqrt{\zeta^2 - 1}} e^{(\omega_n \sqrt{\zeta^2 - 1})t} - \frac{\omega_n}{2\sqrt{\zeta^2 - 1}} e^{-(\omega_n \sqrt{\zeta^2 - 1})t} \right]$$

(iii) $\zeta < 1$

$$x(t) = \frac{\omega_n}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \sin \omega_n t$$

• Step: Unit

$$f(t) = 1 \quad F(s) = 1/s$$

(i) $\zeta = 1$

$$x(t) = 1 - (1 + \omega_n t) e^{-\omega_n t}$$

(ii) $\zeta > 1$

$$x(t) = 1 - e^{-\zeta\omega_n t} \left\{ \frac{\zeta + \sqrt{\zeta^2 - 1}}{2\sqrt{\zeta^2 - 1}} e^{\omega_n \sqrt{\zeta^2 - 1} t} - \frac{\zeta - \sqrt{\zeta^2 - 1}}{2\sqrt{\zeta^2 - 1}} e^{-\omega_n \sqrt{\zeta^2 - 1} t} \right\}$$

(iii) $\zeta < 1$

$$x(t) = 1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n t + \theta)$$

Type of transient response $\rightarrow$ closed loop poles

Shape of transient response $\rightarrow$ closed loop zeroes.

Transient response of an underdamped can be specified by rise time, peak time, settling time and maximum % overshoot:

(i) Rise time: $t_r = \frac{\pi - \phi}{\omega_d}$

(ii) Peak time: $t_p = \frac{\pi}{\omega_d}$

(iii) Maximum Overshoot: $1 + e^{-\frac{\pi \zeta}{\sqrt{1-\zeta^2}}}$

(iv) Settling time:

Amplitude of underdamped response will be always b/w the envelope $1 \pm \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta \omega_n t}$

$MO = e^{-\frac{\pi \zeta}{\sqrt{1-\zeta^2}}}$

It is independent of ω_n

Time constant of these envelope is $\frac{1}{\zeta \omega_n}$

Dominant pole $\rightarrow$ Poles nearer to the origin
Conjugate poles nearer to origin

$\pm 2\%$ of the desired output

This happens when $e^{-\zeta \omega_n t_s} < 0.02$

or, $t_s \approx \frac{4}{\zeta \omega_n}$

If we have real poles and complex poles,
Then, $\frac{\text{Real pole}}{\text{Real part of complex pole}} > 5 \Rightarrow$ Complex poles are dominant

A disturbance signal is an unwanted input signal that affects the system's output.
Feed back loop helps in controlling and eliminating disturbance signals.

If we have a unity feed back, then the actual output becomes equal to the desired output.

$Y(s) = \frac{X(s)}{H(s)}$

To suppress the disturbance signal, $|G, G_1, H| \gg 1$

• Automatic Controllers:

- (i) On-off controllers (ii) proportional controllers (iii) Integral Controllers
- (iv) proportional plus Integral controllers (v) proportional plus differential controllers.
- (vi) proportional plus Integral plus differential controllers.

power sources $\rightarrow$ Electricity / pressurised fluids (air or oil)

Industrial controllers $\rightarrow$ ~~Electrical~~ Electronic, Hydraulic, pneumatic

Flow of steam through nozzle can be considered as an adiabatic expansion.

At the outlet, velocity increases, pressure drops, expands, enthalpy decreases.

This high velocity steam is capable of driving turbine.

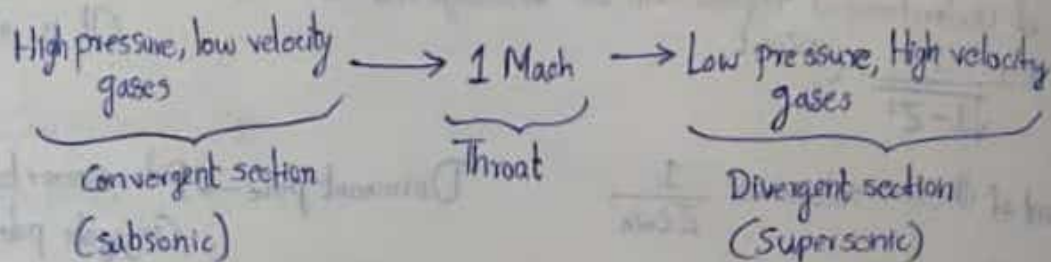
• Types of nozzles:

Convergent-Divergent nozzle is used in the case where the back pressure is less than the critical pressure.

(i) Convergent

(ii) Divergent

(iii) Convergent-Divergent



Friction exists b/w steam and sides of nozzle which produce heat.

Compressible fluid: Density changes with change in pressure.

Gases and Vapours $\rightarrow$ Compressible.

Liquid $\rightarrow$ Incompressible

• Velocity of fluid leaving the nozzle:

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$$

$$\Rightarrow V_2 = \sqrt{2(h_1 - h_2) + V_1^2}$$

h in J/kg and V in m/s

Now, h in kJ/kg

$$\Rightarrow V_2 = \sqrt{2000(h_1 - h_2) + V_1^2}$$

$$V_2 = \sqrt{2000(h_1 - h_2) + V_1^2}$$

• Equation of continuity:

$$m = \frac{AV}{v}$$

$m \rightarrow$ mass of fluid in kg/s

$A \rightarrow$ Cross sectional area in m^2

$V \rightarrow$ Velocity across the section m/s

$v \rightarrow$ Specific volume of the fluid m^3/kg

$$m = \frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2}$$

For steady flow

• Critical Pressure Ratio:

$\frac{P_2}{P_1}$ } pressure ratio at throat

$n = 1.3 \rightarrow$ Super heated steam

$n = 1.35 \rightarrow$ Saturated steam

$n = 1.035 + 0.1x \rightarrow$ Wet steam

$$\frac{P_2}{P_1} = \left(\frac{2}{n+1} \right)^{\frac{n}{n-1}}$$

• Nozzle Efficiency (K)

$$K = \frac{\text{Useful heat drop}}{\text{Isentropic heat drop}}$$

$$0.85 \leq K \leq 0.95$$

• Factors affecting K :

(i) Material of nozzle

(ii) Size & shape of nozzle

(iii) Finish of the nozzle

(iv) Angle of divergence

(v) Nature of fluid and its state

(vi) Friction

(vii) Fluid velocity

(viii) Turbulence in the flow passages.

• Super-saturated flow or Meta stable flow in nozzles:

High velocity steam while passing through throttle $\rightarrow$ Insufficient time for condensation $\rightarrow$ Condensation delayed

Super Saturation.

Super Saturated steam $\rightarrow$ Steam that remains in the wet region without holding any liquid.
 Wilson line $\rightarrow$ Locus of points where condensation occurs. lies b/w 4 and 5% moisture curves in saturation region.

If there is friction, it is not isentropic.

No. of nozzles $\rightarrow$ whole number.

$$8 \text{ bar} \quad 0.8 \quad 721.11 + (0.8 \times 2048.0) = 1638 + 1638.88 = 2359.51 \text{ kJ/kg}$$

$$11 \quad 797.55 \text{ kJ/kg} \quad 1384.55 \text{ kJ/kg} \quad \rightarrow 2047.95 \quad 0.001127 \rightarrow V_f$$

$$762.81 \text{ kJ/kg} \quad 3020 \text{ kJ/kg} \quad \rightarrow 2257.19 \quad 0.19444 \rightarrow y$$

Specific enthalpy to enthalpy $\rightarrow$ Multiply by mass flow rate.

$$\frac{1}{5 \text{ bar}}$$

$$\frac{A}{P} = \frac{C \pm B}{B \pm A} \quad \Leftarrow \quad \frac{C}{B} = \frac{B}{A}$$

$$f(D,E) = \frac{g}{h} (x,y,z) = \frac{g}{h} (x,y,z) + \frac{g}{h} (x,y,z)$$

$$P_k + Q_k = R$$

$$f(x,y,z) = 0 \Rightarrow f(x,y,z) = 0$$

$$\frac{q_x}{P} = \frac{q_y}{R} = \frac{q_z}{S}$$

$$f(x,y,z) = f(x,y,z) = f(x,y,z)$$

example

$$q_x = p_x + v_x q_y$$

$$(iv) \text{ Type II: } (x,y,z)$$

$$Z = p_x + v_x + f(v_x)$$

$$(v) \text{ Type III: } (x,y,z)$$

$$f(v_x, z) = 0$$

$$Z = f(x,y,z) = f(v_x, z)$$

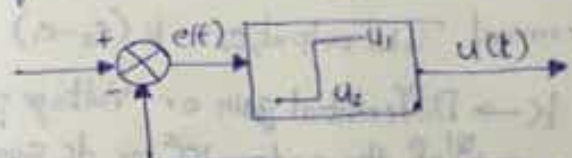
$$\Rightarrow v_x = x + y$$

$$\frac{q_x}{P} = \frac{q_y}{R} = \frac{q_z}{S}$$

- On-off controllers
- Two position control system
- Actuating element has two fixed positions - On & off

$$u(t) = u_1, \text{ for } e(t) > 0 \rightarrow \text{On}$$

$$u_2, \text{ for } e(t) < 0 \rightarrow \text{Off}$$



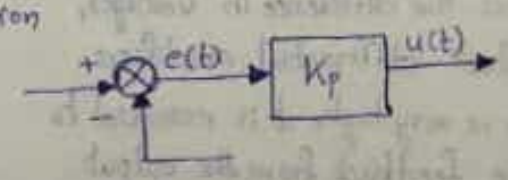
u_1 & $u_2 \rightarrow$ Max and Min positions
 $u(t) \rightarrow$ Output signal $e(t) \rightarrow$ Actuating error signal.

- proportional Controllers: proportional control action

$$u(t) = K_p e(t)$$

$$\Rightarrow \frac{u(s)}{e(s)} = K_p \rightarrow \text{Proportional gain}$$

(This is different from position error constant)



proportional controller is essentially an amplifier with adjustable gain.

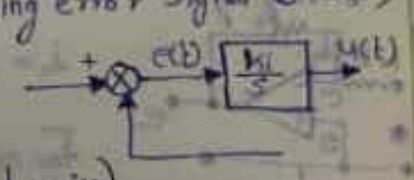
- Integral Controllers: Integral control action

Controller output $u(t)$ is changed at a rate proportional to the actuating error signal $e(t)$

$$\frac{d}{dt} u(t) = K_i e(t) \Rightarrow u(t) = K_i \int e(t) dt$$

$$\frac{u(s)}{e(s)} = \frac{K_i}{s}$$

$K_i \rightarrow$ Adjustable constant (Integral gain)



- proportional plus Integral controller (PI)

$$u(t) = K_p e(t) + K_i \int e(t) dt \Rightarrow \frac{u(s)}{e(s)} = K_p + \frac{K_i}{s} = K_p \left(1 + \frac{1}{T_i s} \right)$$

$$K_i = \frac{K_p}{T_i}, T_i \rightarrow \text{Integral time.}$$

- proportional plus derivative Controller (PD)

$$u(t) = K_p e(t) + K_d \frac{d}{dt} e(t)$$

$K_d =$ Derivative gain

$$\frac{u(s)}{e(s)} = K_p + K_d s = K_p (1 + T_d s) \quad T_d \Rightarrow \text{Derivative time}$$

$$T_d = \frac{K_d}{K_p}$$

- proportional plus Integral plus derivative (PID) Controller:

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{d}{dt} e(t)$$

$$\frac{u(s)}{e(s)} = K_p + \frac{K_i}{s} + K_d s = K_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$

We can get control actions electronically using Operational amplifiers (Op amp). They are used to amplify signals in sensor circuits. Also used in filters used for compensation purpose. Input voltages $\rightarrow e_1$ and e_2 Total Input = $e_2 - e_1$



$e_1 \rightarrow$ Inverted at -ve terminal
 $e_2 \rightarrow$ Not inverted

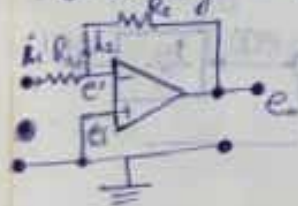
Total output, $e_0 = K(e_2 - e_1)$

$K \rightarrow$ Differential gain or Voltage gain
 $\rightarrow |K|$ of the order 10^5 for dc signals
 $\rightarrow$ low frequency (< 10 Hz) ac signals
 $\rightarrow$ Value of K decreases with increase in frequency
 $\rightarrow K=1$ (approx) for frequencies above 10 Hz

Op amp amplifies the difference in voltages, it is also called differential amplifiers. Since the gain is very high, it is essential to have a negative feedback from the output to the input to make the amplifier stable.

In ideal op amp $\rightarrow$ Input impedance is ∞ & output impedance is '0'. No current flows into the input and output voltage is not affected by the load connected to the output terminal.

• Inverting Amplifier:



$$i_1 = \frac{e_1 - e'}{R_1} \quad \text{and} \quad i_2 = \frac{e' - e_0}{R_2}$$

For an ideal op amp, negligible current flows into it $\Rightarrow i_1 = i_2$

$\Rightarrow K$ is very large, e' very small $\rightarrow e' \approx 0$

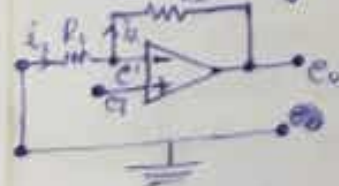
$$K = \frac{R_2}{R_1}$$

$$\Rightarrow \frac{e_1}{R_1} = -\frac{e_0}{R_2} \Rightarrow e_0 = -\frac{R_2}{R_1} e_1$$

If $R_1 = R_2$, the op amp circuit acts as an inverter.

Output is negative.

• Non-inverting Amplifier:



$$i_1 = \frac{0 - e'}{R_1}, \quad i_2 = \frac{e' - e_0}{R_2} \Rightarrow i_1 = i_2$$

$$\Rightarrow -R_2 e' = R_1 (e' - e_0) \Rightarrow (R_1 + R_2) e' = R_1 e_0$$

$$\Rightarrow e' = \frac{R_1}{R_1 + R_2} e_0 \quad \text{but, } e_0 = K(e_2 - e_1) = K(e_1 - e')$$

$$= K \left(e_1 - \frac{R_1}{R_1 + R_2} e_0 \right)$$

$$\Rightarrow e_0 \left(\frac{1}{K} + \frac{R_1}{R_1 + R_2} \right) = e_1 \quad \frac{1}{K} \text{ can be neglected}$$

$$\Rightarrow e_0 = \left(1 + \frac{R_2}{R_1} \right) e_1$$

(iii) R and C in series:

$$Z(s) = \frac{1 + RCs}{C}$$

(i) R connected across V_i ; $Z(s) = R$

(ii) C connected across V_i ; $Z(s) = \frac{1}{Cs}$

(iv) R and C in parallel:

$$Z(s) = \frac{R}{1 + RCs}$$

(i) Proportional Controller:

$$K_p = \frac{R_4 R_2}{R_3 R_1}$$

(ii) Integral Controller:

$$\frac{K_i}{s} = \frac{R_4}{R_3 R_1 C}$$

(iii) PI Controller:

$$\frac{E_o(s)}{E_i(s)} = K_p + \frac{K_i}{s}$$

(iv) PD Controller:

$$\frac{E_o(s)}{E_i(s)} = K_p + K_d s = \frac{R_4 R_2}{R_3 R_1} + \frac{R_4 R_2 C s}{R_3}$$

(v) PID Controller:

$$\frac{E_o(s)}{E_i(s)} = K_p + \frac{K_i}{s} + K_d s = \frac{R_4 R_2}{R_3 R_1} + \frac{R_4}{R_3 R_1 C s} + \frac{R_4 R_2 C s}{R_3}$$

• Steady State Error of feedback control ($t \rightarrow \infty$)

To reduce steady state error, feed back loops are introduced.

$G(0) \rightarrow$ dc gain (> 1)

$$1 - G(0) > \frac{1}{1 + G(0)}$$

$G(s) \rightarrow$ Open loop transfer function.

$$SS(dp) > SS(clp)$$

The number of s in the denominator indicates number of integrations. The number of integrations N is often called type number.

(i) Step input: Constant

$$e_{ss} = \frac{A}{1 + K_p}$$

$K_p = \lim_{s \rightarrow 0} G(s) H(s)$ is called position error constant.

If $N \geq 1$, $K_p = \infty$ and $e_{ss} = 0$

(ii) Ramp Input: linear

$$e_{ss} = \frac{A}{K_v}$$

$$K_v = \lim_{s \rightarrow 0} s G(s) H(s)$$

Velocity error constant.

Type 0 $\rightarrow K_v = 0 \Rightarrow e_{ss} = \infty$

Type 1 $\rightarrow K_v = \text{cons} \Rightarrow e_{ss} = \frac{A}{K_v}$

Type 2 or higher $\rightarrow K_v = \infty \Rightarrow e_{ss} = 0$

The error constants K_p , K_v and K_a of a control system describe the ability of a system to reduce or eliminate the steady state error. They are used as numerical measures of steady state performance. By increasing the error constants, we can minimise the steady state error.

(iii) Acceleration Input: Parabolic

$$e_{ss} = \frac{A}{K_a} \quad K_a = \lim_{s \rightarrow 0} s^2 G(s) H(s) \quad \text{Acceleration error constant.}$$

Type 0 and 1 $\rightarrow K_a = 0 \Rightarrow e_{ss} = \infty$

Type 2 & above $\rightarrow K_a = \infty \Rightarrow e_{ss} = 0$

Type 2 $\rightarrow K_a = \text{cons} \Rightarrow e_{ss} = A/K_a$

Modern - Control Theory $\rightarrow$ Time-domain approach $\rightarrow$
Classical - Control Theory $\rightarrow$ Complex = frequency-domain approach

• Stability:

Design $\rightarrow$ Transient response, steady state error and stability.

Total response = Natural response + Forced response.

Linear-time-invariant system.

- i) Stable if natural response $\rightarrow 0$ when $t \rightarrow \infty$
- ii) Unstable if natural response grows unbounded when $t \rightarrow \infty$
- iii) Marginally stable if natural response remain constant or oscillates as $t \rightarrow \infty$.
(neither decays nor grows)

Stable $\rightarrow$ Closed loop poles located on the left half of the s-plane.
Negative real part of poles.
Have exponential decay term $e^{-\lambda t}$

Unstable $\rightarrow$ Any of the closed loop poles on the right side of s-plane.
Imaginary axis poles of multiplicity greater than one.
leads to sum of responses as $Ae^{n\omega t} \sin(\omega t + \phi)$; $n = 1, 2, 3, \dots$ approaches ∞ when $t \rightarrow \infty$

Marginally Stable $\rightarrow$ Imaginary axis poles of multiplicity one yields pure sinusoidal oscillations

• Routh - Hurwitz's Stability Criterion:

It helps us to find the number of closed loop poles that lie in the right half s-plane without factorising the denominator.

(i) $a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0 = 0$

Coefficients are real quantities and $a_0 \neq 0$
possibility of any zero root is eliminated.

- (ii) Any of the coefficients are zero or negative,
Root or roots are imaginary or that have +ve real parts.
System is not stable.

- (iii) If all coefficients are +ve, arrange them in rows and columns.

It states that, "the number of roots with positive real parts is equal to the number of changes in sign of the coefficients of the first column, of the Routh array."

• Special Conditions:

(i) Zero in the first column of a row

- we consider a small quantity ϵ in that particular cell.
- proceed as usual.
- Giving conditions as $\epsilon \rightarrow 0^+$ and $\epsilon \rightarrow 0^-$
- Then observing sign changes.

(ii) Zeros in the entire row:

- Entire row of zeroes.
- Row previous to row of Zero $\rightarrow$ Even polynomial / Auxiliary polynomial $p(s)$.
- Even polynomial will have only even powers of s .
- They will have only roots that are symmetrical about the origin.
- Symmetry can occur under 3 conditions: (a) Symmetrical and real.
- $\frac{d}{ds} p(s)$ (b) Symmetrical and imaginary.
- (c) Quadrantal
- Replace the row of zeros by coefficients of $\frac{d}{ds} p(s)$
- proceed as usual.
- If there is no sign change under the even polynomial coefficients in the first column, roots will be on the imaginary axis with multiplicity one. Then it is marginally stable.
- If there is sign change under the even polynomial, then we have roots on the RHP, system is unstable.

• Root Locus Method:

- Graphical representation of closed loop poles as a system parameter is varied.
- Powerful method for analysis of stability and transient response.

Closed loop transfer function, $T(s) = \frac{G(s)}{1 + G(s)H(s)}$

Characteristic eqn $\rightarrow 1 + G(s)H(s) = 0 \Rightarrow G(s)H(s) = -1 + j0$

$$\Rightarrow |G(s)H(s)| = 1 \quad \text{Magnitude Condition}$$

$$\Rightarrow \angle G(s)H(s) = \pm (2k+1)\pi \quad \text{Angle Condition}$$

Odd multiples of 180° , $k = 0, 1, 2, \dots$

The values of s that satisfies the magnitude and angle condition are the roots of the characteristic eqn or the closed loop poles.

In most cases there is a gain $K \rightarrow T(s) = \frac{K G(s)}{1 + K G(s)H(s)}$

$$\Rightarrow 1 + K G(s) H(s) = 0 \quad \text{Characteristic eqn}$$

The locus of roots of the characteristic eqn when the parameter K is varied from zero to infinity is known as root locus.

• Root Locus Procedure:

(i) Step 1: $1 + K G(s) H(s) = 0$ } characteristic eqn

$$1 + K \frac{\prod_{i=1}^m (s + z_i)}{\prod_{j=1}^n (s + p_j)} = 0 \Rightarrow K = - \frac{\prod_{j=1}^n (s + p_j)}{\prod_{i=1}^m (s + z_i)}$$

(ii) Step 2: K varies from $K=0$ to $K=\infty$

$$K=0 \Rightarrow \prod_{j=1}^n (s + p_j) = 0 \Rightarrow \text{Roots are simply the poles of } G(s) H(s)$$

$$K=\infty \Rightarrow \prod_{i=1}^m (s + z_i) = 0 \Rightarrow \text{Roots are simply the zeroes of } G(s) H(s)$$

$\therefore$ Root locus starts from the poles and ends at the zeroes of $G(s) H(s)$ as K varies from 0 to ∞

n = Number of poles and m = Number of zeroes ($n > m$)

$\Rightarrow (n-m)$ zeroes lie at infinity

$\Rightarrow (n-m)$ branches of root loci will go to zeroes at infinity.

(iii) Step 3:

Locate the segments of the real axis that are part of the root loci.

Any point on the root loci satisfy the angle condition.

The real axis segment to the left of odd number of poles and/or zeroes will be part of the root locus.

(iv) Step 4: Determine the separate loci (SL)

Number of separate loci = Number of poles = n

(v) Step 5: Root loci must be symmetrical with respect to the real axis because the complex poles/roots occur as conjugate pairs.

(vi) Step 6: The root loci proceed to zeroes at infinity along the asymptotes centered at σ_A and with angles ϕ_A

$$\sigma_A = \frac{\sum \text{poles} - \sum \text{zeroes}}{n-m}$$

$$\phi_A = \frac{(2k+1)180^\circ}{n-m}; k = 0, 1, 2, \dots$$

(vii) Step 7: Determine the break in point/break away point.

There will be multiplicity of roots at break away point.

To find break away point: $K = P(s)$ $P(s) \rightarrow$ characteristic eqn in reverse.

$\frac{dK}{ds} = 0$ will give break away point.

(viii) Step 8: Determine the points where root loci crosses the imaginary axis.

This can be found out by Routh's Stability Criterion.

Find the value of K .

$$1 + K G(s) H(s) = 0$$

Write the even polynomial and find the value of S .

The root loci crosses the imaginary axis at this complex conjugate of S .

(ix) Step 9: Determine the angle of departure from a complex pole and the angle of arrival of the locus at a complex zero, using the angle criterion.

(x) Step 10: Choose the test points in such a way that they satisfy the angle criterion. Draw the loci passing through these points.

(xi) Step 11: Determine the pair of dominant complex conjugate poles such that the damping ratio is 0.5

• Design with root locus:

$$P_s = -\zeta\omega_n \pm j\omega_d$$

• Signal Flow Graph:

They are alternative to block diagrams.

It consists of branches, nodes.

Branches $\rightarrow$ System (Transfer functions) with arrows.

Nodes $\rightarrow$ Signals

Signal flow graphs are graphical representation of linear equations.

Transmittance $\rightarrow$ Gain acquired while travelling from one node to another.

• Armature Controlled dc motor:

$$V_a = I_a R_a + L_a \frac{di_a}{dt} + e_b, \quad T = K_t i_a, \quad T = J \frac{d\omega}{dt} + B\omega, \quad e_b = K_b \omega, \quad \omega = \frac{d\theta}{dt}$$

• Mason's Gain Formula:

$$\Delta = 1 - (\sum ILG) + (\sum G_1 P_1 T_1 N_1 T_1 L_1)$$

$$T(s) = \frac{1}{\Delta} \sum_k P_k \Delta_k$$

(i) Forward path gains (FPG)

(ii) Individual loop gains (ILG)

(iii) Gain products of two non-touching loops. (G₁P₁T₁N₁T₁L₁)

(iv) Calculation of Δ and Δ_k

(v) Transfer function

• Position Control Servo System:

A Servo System is an ~~analog~~ automatic device that uses error-sensing negative feedback to correct the action of the mechanism.

→ It is a pair of potentiometers which convert the input and output position into proportional voltages.

→ The desired output position is given as the input to the first potentiometer.

→ The electric potential of the arm is proportional to the angular position 'r' of the arm.

→ The output shaft position determines the angular position 'c' of the arm of the output potentiometer.

→ e → Difference b/w the input angular position 'r' and output angular position 'c' is the error signal (e).

$$e = r - c$$

Input voltage corresponding to 'r' → $e_r \Rightarrow e_r = K_r r$

Output voltage corresponding to 'c' → $e_c \Rightarrow e_c = K_c c$

$K_o \rightarrow$ Proportionality constant

$$\text{Error voltage, } e_v = e_r - e_c$$

The error voltage e_v is applied to the amplifier with gain K_i . Then output voltage of amplifier

$$e_a = K_i e_v$$

- This output voltage of amplifier is applied to circuit of dc motor.
- A fixed voltage is applied to field winding.
- When there is an error, motor rotates to nullify the error.
- If field current is constant, torque developed by the motor $\propto$ Armature Current ' i_a '

$$T = K_t i_a \quad K_t \rightarrow \text{Motor Constant}$$

- When armature is rotating, a voltage proportional to the product of flux and angular velocity is induced in the armature.
- For a constant flux, voltage produced is, back emf,

$$e_b = K_b \frac{d\theta}{dt} = K_b \omega$$

$\theta \rightarrow$ Angular displacement of motor shaft.

$K_b \rightarrow$ Back emf constant of motor.

- The speed of armature controlled dc servomotor is determined by the armature voltage e_a/v_a
- Diff eqn of armature circuit is,

$$v_a = i_a R_a + L_a \frac{di_a}{dt} + e_b = i_a R_a + L_a \frac{di_a}{dt} + K_b \frac{d\theta}{dt}$$

$$J_o \ddot{\theta} = T - b_o \dot{\theta} \Rightarrow J_o \ddot{\theta} + b_o \dot{\theta} = T = K_t i_a$$

$J_o \rightarrow$ Mass moment of inertia of the motor, load and gear train with respect to the motor shaft axis

$b_o \rightarrow$ Viscous friction coefficient of the combination of the motor, load and gear train referred to motor axis.

$$T(s) = \frac{k}{Js^2 + Bs + K}$$

• Design with root Locus:

Compensators are additional components in a system. It will introduce new poles / zeroes into the openloop transferfunction of the system.

Poles Introduced $\rightarrow$ pulls root locus to the right and reduce relative stability and increase settling time

Zeroes Introduced $\rightarrow$ pulls root locus to the left and increase the relative stability and decrease settling time.

Lead Compensator $\rightarrow$ Improves only transient response. The transfer function of lead compensator will have a compensating zero and compensating pole.

Compensator zero @ real axis $\rightarrow$ Improve transient response $\rightarrow$ Amplify high frequency noise

Negligible effect on root locus in the region where two dominant closed loop poles are located. $\leftarrow$ @ real axis $\leftarrow$ Eliminated by compensating pole.

C-pole far away from C-zero $\rightarrow$ Not effective in suppressing the noise. } To avoid this, pole is located 3 to 10 times the value of zero location.
C-pole too close to C-zero $\rightarrow$ Will not allow zero to do its job.

Lag Compensator $\rightarrow$ When there is satisfactory transient response but unsatisfactory steady state response i.e. error requirement.
A compensating zero and compensating pole is introduced.

To preserve transient response $\rightarrow$ C-pole and C-zero should have negligible effect on shape of root locus. So, they are placed very close to each other. If they are very close to each other at the origin, it will reduce steady state error as the error constant increases.

Lag-Lead Compensator $\rightarrow$ When both transient and steady state responses are not satisfactory.

• Frequency Response:

The steady state response of a system to a sinusoidal input is known as frequency response. Sinusoidal input to a linear system gives a sinusoidal output of the same frequency. But the amplitude and phase of the output may differ from input. These differences will be functions of frequency.

$$G(s) = \frac{1}{1+Ts} \Rightarrow \text{For RC circuit} \Rightarrow G(s) = \frac{1}{1+RCs} \quad \left. \vphantom{G(s)} \right\} \text{First Order}$$

$$\text{Time constant, } T = RC \quad e_i = A \sin \omega t$$

$$E_i(s) = \frac{A\omega}{s^2 + \omega^2} \quad e_o(t) = \frac{A}{\sqrt{1+\omega^2 T^2}} \sin(\omega t + \phi)$$

$$\text{Magnitude of frequency response} = \frac{A}{\sqrt{1+\omega^2 T^2}} = M$$

$$\text{Phase angle, } \phi = \tan^{-1} \left(\frac{-\omega T}{1} \right) = \tan^{-1}(-\omega T) = -\tan^{-1}(\omega T)$$

For a complex frequency, we get these same results.

Frequency response, $e_o(t) = A |G(j\omega)| \sin(\omega t + \angle G(j\omega))$

Magnitude of output, $M_o = M_i \cdot M$

phase angle, $\phi_o = \phi_i + \theta$

$G(j\omega) = \frac{1}{1 + j\omega T}$ $\rightarrow$ Sinusoidal Transfer function

Bode Diagram:

Consists of two graphs:

(i) $\log |G(j\omega)|$ vs ω on a logarithmic scale

(ii) ϕ vs ω on a logarithmic scale.

$$\log |G(j\omega)| = 20 \log_{10} |G(j\omega)| \text{ dB}$$

It is plotted on a semi-log graph.

Y-axis $\rightarrow$ Linear Scale $\rightarrow$ To represent magnitude in dB or phase angles in degree

X-axis $\rightarrow$ Logarithmic Scale $\rightarrow$ To represent frequency range.

When we consider transfer function of $G(s)H(s)$ or $G(j\omega)H(j\omega)$, following factors occur frequently:

(i) Gain, K (ii) Integral term $\frac{1}{s}$ and derivative factors s

(iii) First order factors $(s+a)$ or $(1+Ts)$ in Num or den

(iv) Quadratic factors $(s^2 + 2\zeta\omega_n s + \omega_n^2)$ in Num or den